# DEBASISH DALUI

*Prosper, TX, 75078 | (216) 835 2902*

debasishdalui@gmail.com | *https://www.linkedin.com/in/debasish-dalui/*


An ardent technologist with over two decades of experience, enthusiastic about solving complex business challenges, and instrumental in global enterprise solutions architecture focused on customer adoption across multiple domains (*e.g., banking and financial, retail, oil & gas, manufacturing, government*) across geographies, earning exceptional customer experience, trust, and making an impact. My professional value boils down to the following:

- **Expert in digital transformation, legacy modernization, and cloud (*AWS, Azure, GCP*) adoption**, leveraging Generative AI and large language models to enhance enterprise solutions, ensuring technical alignment with customer success and project timelines.
- **Proficient in designing and integrating scalable complex systems, Cloud-native, low-code no-code applications** (*Business Process and Decision Mngt. System*) ensuring alignment between IT infrastructure and business processes.
- **Extensive experience in IT advisory and architecture workshop execution**; skilled in assessing as-is, conducting gap analysis, defining to-be state, and revamping roadmaps to drive organizational buy-in.
- **Proficient in** API economy, API governance, Domain-Driven Design, Microservices/SOA patterns, Generative AI and Agent, Prompt Engineering, DevSecOps, BPMN, DMN, Cloud Well-Architected frameworks, and TOGAF methodologies**.**
- **Strong leadership qualities with a motivational and dedicated approach**. A trusted advisor and mentor to hundreds of engineers and architects, fostering talent development and partnership-building across global programs.

## PROFESSIONAL EXPERIENCE

**Managing Principal, Enterprise Architecture, LTIMindtree** **Aug 2024 – Present**

Below are my key accomplishments

- Spearheaded scalable application designs for legacy modernization, leveraging cutting-edge technologies to address critical customer challenges, ensuring 100% seamless integration and strong stakeholder alignment.
- Designed cloud-native solutions, applying strangler migration patterns to gradually replace legacy systems, reducing technical debt by 40%, enhancing system performance by 35%, and improving scalability, fault tolerance, and user experience.
- Led the development and delivery of as-is and to-be architecture specifications, driving proof of concepts (PoCs) that accelerated reference architecture adoption by 30%, ensuring high availability and resilience for next-gen IT solutions.
- Provided expert guidance and live demos on Generative AI, Prompt Engineering, LLM, RAG, and Agentic AI, optimizing SDLC efficiency, reducing development efforts by 30-35%, and improving time-to-market and automation capabilities.
- Led cross-functional teams to define core solution tenets, assess modernization challenges, and align with enterprise architecture governance frameworks, driving a 25% reduction in modernization costs.
- Partnered with engineers and architects to develop innovative PoCs, Points of View (PoVs), and prototypes, showcasing interoperability and extensibility, increasing customer engagement by 30%.
- Led enterprise architecture solutions for Fortune 500 oil and gas and manufacturing clients, strengthening pre-sales initiatives, driving a 20-25% increase in deal conversions, and reinforcing technology alignment with strategic business outcomes.
- Developed market-leading sales collateral, amplifying awareness of technology strategy and architecture consulting, contributing to a 15% growth in service adoption, and ensuring scalability and adaptability of enterprise solutions.

**Principal, Enterprise Applications, Infosys** **Sep 2017 – Aug2024**

Below are my key accomplishments

- Led digital transformation and legacy modernization initiatives for top U.S. financial, retail, and government organizations, ensuring 95% alignment with strategic goals while improving scalability and reliability of enterprise systems.
- Advised enterprises on COP implementation, API/microservice/cloud governance, and maturity models, optimizing service lifecycle efficiency by 40%, enhancing resilience, fault tolerance, and interoperability across architectures.
- Facilitated architecture workshops, collaborating with leadership to prioritize initiatives, improve efficiency by 30%, enhance system scalability, and drive innovation through PoCs, PoVs, and prototypes that focused on performance optimization.
- Partnered with cross-functional teams to integrate solutions, providing technical guidance on automation, orchestration, and observability, optimizing system reliability to 99.9% uptime while enhancing fault tolerance and disaster recovery capabilities.
- Collaborated with security teams to define compliance standards, ensuring 100% adherence to data privacy regulations, reinforcing zero-trust architecture principles, and fortifying enterprise security posture.
- Conducted pre-sales architecture workshops on API lifecycle, governance, and cloud-native adoption, driving a 20-25% revenue increase, while ensuring solutions aligned with high availability and robust security standards.

**Principal Architect, Cognizant** **Mar 2005 - Sep 2017**

Below were my accomplishments during my 12.5 years with different customer facing roles.

- Led global banking and financial projects focused on solution architecture, digital transformation, and cloud-native integration, ensuring highly scalable and resilient enterprise systems, delivering projects 100% on schedule and within budget while maintaining 95% alignment with strategic goals.
- Partnered with business and IT leaders to translate technical concepts into strategic enterprise solutions, optimizing architecture reliability, system interoperability, and long-term sustainability, ensuring seamless alignment with business objectives.
- Managed a team of 10+ architects across US-based accounts, driving 75% integration and delivery while generating 25% new opportunities in pre-sales activities, ensuring architectural governance, clear technology roadmaps, and alignment with enterprise-wide strategies.
- Accelerated digital transformation by 30-35% through PoCs, MVPs, and open-source assets, leveraging technical documentation and ROI analysis to enhance architecture scalability, flexibility, and maintainability, ensuring future-proof technology adoption.
- Defined global technical requirements and architecture standards, enhancing security, performance optimization, and cost-effectiveness while establishing robust reference architectures for reliability, fault tolerance, and disaster recovery capabilities.
- Provided expert architectural guidance, launching the Enterprise Architecture Review Board, improving governance, standardization, and collaboration on future enterprise technology strategies and compliance frameworks.

## PRE-SALES EXPERIENCE:

- Over the past 12 years, led pre-sales architectural activities across Fortune 500 banks, financial institutions, insurance companies, oil and gas corporations, manufacturing enterprises, and government agencies, influencing $500M+ in technology investments and driving 20-25% higher deal conversions.
- Collaborated with Exxon Mobil, Halliburton, Shell Oil & Gas, Air Products, Otis, LUCID, Lifelabs, CVS, Recuro Health, CapitalOne, JPMC, Bank of America, World Pay, Citizens Bank, HSBC, National Archives (NARA), Nationwide Insurance, focusing on legacy system modernization, cloud adoption, and cloud-native architecture development, achieving 40% reduction in technical debt and 30-35% improvement in system scalability and performance.

Established maturity models, competency centers, and federated governance frameworks, strengthening technology alignment with business objectives, enhancing interoperability, and ensuring 100% compliance with industry regulations.

## EARLIER CAREER

**Software Engineer** May 2004 - Feb 2005

Satyam Computer Services Ltd., Hyderabad

In my limited journey at Satyam below are my achievements,

- Involved in designing and integrating an Automated and Dynamic Appraisal System using Java, J2EE, ILOG 4.5 Rule Engine, Oracle DB, and IBM WebSphere.

**Software Developer** Jan 2002 - May 2004

Nisus Technologies Pvt. Ltd., Secunderabad

My career started as Software Developer for an emerging and innovative product-suite called "Ubiquity" by Nisus (www.nisusinc.com) as a product development company. Below are my accomplishments,

- Developed a Rule-based expert/Artificial Intelligence (AI) system utilizing JESS language (an Expert System Shell written by Sandia National Laboratories, USA), Java, Swing, XML, XSLT, Oracle, MS-SQL, DB2.
- Developed a customized scalable Compliance Server developed using Java 2.

## SKILLS

- **Enterprise Solution Architecture:** Application Design, Enterprise Solutions Architecture, Platform Engineering, J2EE, SOA, BPMN 2.0. DMN, Object-oriented Design, Distributed Systems, Middleware, TOGAF, Abacus, Archimate, Zachman, LeanIX, Salesforce, Design pattern, Anti-pattern,
- **Cloud Technologies:** Cloud-native/agnostic/hybrid Platform Engineering, Cloud Infrastructure, SaaS, IaaS, PaaS. Amazon Web Services (AWS), Microsoft Azure, Google Cloud (GCP), PCF (Pivotal Cloud Foundry), J2EE, SOA, Distributed Systems, Middleware, TOGAF, Design pattern, Anti-pattern
- **Enterprise Integration and API Management:** API economy, Microservices, Web Services (SOAP, RESTful), Service design, IBM API Connect, Salesforce MuleSoft API Platform, API Gateway (AWS, Apigee, Akana), Appian, Camunda, JBPM, Kafka, IBM Blueworkslive, IBM BPM/Lombardi, Decision Center, Decision Server, IBM Business Rules Engine (BRE), IBM ODOM, Drools
- **Container/Application servers/Serverless:** Kubernetes, AWS ECS, EKS, AWS Lambda, Fargate, Azure Functions, WebSphere, Weblogic, JBoss server, ESB
- **SDLC and DevOps Practices:** Agile development, Scrum and Kanban methodology, PI Planning, DevOps, DevSecOps, CI/CD, TDD, BDD, GitHub, Jenkins, Terraform, CircleCI, Veracode, Static Code Analysis, SonarQube

- **Data Architecture and Analytics:** Data Structure, Data governance, Data privacy, Data security, and Data quality management, Database (AWS RDS, MS SQL Server 2014, Oracle, SQL Server, Postgres, Mongo, Cassandra, Dynamo DB, Cosmos DB) Big Data platforms (Hadoop, Spark), and Data warehousing solutions (AWS Redshift)
- **Emerging Technologies and Innovation:** Generative AI (Amazon Bedrock, Azure OpenAI), Artificial Intelligence (AI) and Machine Learning (ML), Blockchain, Internet of Things (IoT), Edge Computing, Natural Language Processing (NLP)
- **Specifications/Modeling:** UML, BPMN, DMN, Open Bank Project, BIAN, Swagger specification
- **Programming languages:** Java, JavaScript, Python, Node.js, C#, JESS (by Sandia National Labs)
- **UI/UX technology:** JSP, JavaScript, Angular, React, React-native

## KEY ROJECT EXPERIENCE:

1. **"LFS 2.0" LUCID Financia Services Modernization**
    - **Client: LUCIDMOTORS**
    - **Role:** Solutions Architect
    - **Technologies Used:** Microservices, Java Springboot, Drools Rule Engine, DMN, AWS, EKS, ECS, Lambda, Postgres, Terraform, Generative AI/Github-Copilot
    - **Project Role Description:** LUCID financial services is looking to transform the current Financial Platform part of a Technology revamp. Financial Platform redesign will result in a Mobile first customer experience with reduces maintenance & operational costs.
    - **Key contributions:**
        - Designing a “micro-app” based architecture to support a reusable data model, standardized APIs, and ensure performance, scalability, stability, and maintainability.
        - Refactor Node.js services into Java Spring boot microservices and Domain-driven design; migrate MongoDB to PostgreSQL to align technical debt.
        - Identified and leveraged Github-Copilot in key SDLC phase to improve quality, enhance efficiency and reduce overall effort around 40-50%.
        - Led MVP using strangler migration patterns to gradually replace legacy systems.
        - Incorporate the Drools decision engine to externalize the business rules, leveraging OMG DMN standard template and a microservice based execution approach for enhanced management.
        - Collaborated on integration solutions, conducting reviews with CTOs and architects.

2. **"MyWerk" Platform Modernization Assessment, OTIS**
    - **Client: OTIS**
    - **Role:** Solutions Architect
    - **Technologies Used:** Microservices, .Net, Azure AKS, Azure SQL MI, Airflow, Terraform, Generative AI/Copilot
    - **Project Role Description:** Design and cloud-native architecture for a global elevator and escalator manufacturer, modernizing the 'MyWerk' platform leveraging Azure hyperscaler, Generative AI to expedite SDLC phases and collaborating with stakeholders to ensure scalable, high-performance solutions.
    - **Key contributions:**
        - Diagnosed root causes of issues in top services handling 45-65 million daily requests.
        - Designed cloud-native architecture with deployable data pipeline and microservices on Azure.
        - Identified the scope of Generative AI to expedite SDLC by 30-40%.
        - Collaborated on integration solutions, conducting reviews with CTOs and architects.
        - Led MVP using strangler migration patterns to gradually replace legacy systems.

3. **Service Management System (SMS 8.1) Platform modernization**:
    - **Client: OTIS**
    - **Role:** Solutions Architect
    - **Technologies Used:** React, API, Microservices, BFF pattern, .Net, Azure AKS, Azure SQL MI, Azure Functions, Terraform, Generative AI/Copilot

- **Project Role Description:** Re-architecture and modernization of a global service management system for a leading elevator and escalator manufacturer, enhancing integration, user experience, and scalability across EMEA, APAC, and China regions.
- **Key contributions:**
    a. Defined the cloud native architecture utilizing React, API and microservices, BFF layer pattern deployable to Azure hyperscale.
    b. Understand and document the scope of using Generative AI that expedites the SDLC phases by roughly 30%.
    c. Collaborated with cross-team stakeholders, architects, and leads to design integration solutions that align with overall business goals and solution reviews with CTOs and architects.
    d. Applied the strangler migration patterns to replace legacy system with reduced risk gradually.
    e. Defined the CICD standards to align with multiple countries for zero-downtime deployment.

4. **Retirement Plan Platform Modernization**:
- **Client: Vanguard**
- **Role:** Enterprise Solution Architect
- **Technologies Used:** Angular 17, Appian BPM, Postgres DB, Microservices (Spring-boot), API, REST, Drupal, Redshift
- **Project Role Description:** Vanguard aims to digitally transform and enhance its full-service defined contribution (DC) business, revolutionizing the corporate retirement plan experience for sponsors and participants.
- **Key contributions:**
    a. Designing the headless low-code/no-code solution involved the workflow automation, case management with Appian BPM and microservice cloud-native.
    b. Designed and led the modernization of legacy DC systems from the mainframe.
    c. Applied the 12-factor principal, Microservice patterns like SAGA, choreography. Circuit-breaker, CQRS, micro-DB pattern, Domain Driven Design etc.
    d. Collaborated with cross team stakeholders, product teams, architects and leads to design integration solutions that align with overall business goals and solution review client leadership and architects.
    e. Trained 25+ engineers on relevant new technologies, preparing them for upcoming projects.

5. **National Archive Public Search Application (https://catalog.archives.gov/) Modernization**:
- **Client: National Archives and Records Administration (NARA), Federal Govt.**
- **Role:** Enterprise Solution Architect
- **Technologies Used:** AWS Cloud services, AWS OpenSearch, IIIF Cantaloupe Server, S3, DynamoDB, Lambda, Fargate backed by EKS, EC2, MySQL, AWS RedShift, AWS Airflow, AWS Textract, React, Node.js, and Splunk.
- **Project Role Description:** The goal of enhancing the National Archives and Records Administration (NARA) was to modernize its applications (https://catalog.archives.gov/), ensuring greater efficiency, scalability, and accessibility for users; I was responsible for defining the complete architecture and enabling efficient searches across petabyte-scale archival data in the AWS environment.
- **Key contributions:**
    a. Designed a cloud-native architecture leveraging AWS managed services such as OpenSearch for the core search engine, Airflow for ingesting terabytes of data weekly, and AWS Textract for extracting information from various images and file systems.
    b. Applied the AWS Well-Architected frameworks, 12-factor principles and microservice patterns, including SAGA, choreography, circuit-breaker, CQRS, and shared database pattern, Domain Driven Design etc. that helped to improve the application overall performance as per client approved metrics.
    c. Developed a GitHub branching strategy for both regular development and maintenance teams.
    d. Created Terraform scripts to deploy infrastructure as code seamlessly.

    e. Designed a DevOps pipeline using CircleCI and GitLab.

6. **DMV Ecosystem Modernization**:
   - **Client: New York DMV**
   - **Role:** Enterprise Solution Architect
   - **Technologies Used:** React, Spring-Boot, AWS Gateway, API, Microservices, AWS, S3, Postgres DB, SQS, SNS.
   - **Project Role Description:** A US State DMV aimed to modernize its Driving Licensing System ecosystem to enhance customer service, transaction processing, self-service, and efficiency across 400+ agencies using a modern communication approach. As an enterprise solution architect, I have defined the services utilizing APIs, Microservices, SOAP/Web-based, and message-based to exchange data seamlessly with agencies.
   - **Key contributions:**
     a. Designed a cloud-native architecture utilizing APIs, microservices, SOAP-based services, and message queue implementations to connect with over 400 agencies through their provided channels.
     b. Applied the 12-factor principles along with microservice and SOA patterns, Behavior Driven Development (BDD) approach.
     c. Collaborated with cross-team stakeholders, including product owners and architects, to design integration solutions that align with overall business goals.
     d. Conducted regular manual code reviews to ensure adherence to DMV shared guidelines.
     e. Trained engineers as necessary in relevant technologies.

7. **Center of Process (COP) - API/Microservice/Multi-Cloud**:
   - **Client: HSBC**
   - **Role:** Enterprise Solution Architect
   - **Technologies Used:** API, Microservices, Design pattern, Anti-pattern, AWS and GCP services.
   - **Project Role Description:** Advisory on "Center of Process (COP) - API/Microservice/Multi-Cloud" setup for one of largest global bank, which includes as-is assessment, gap, roadmap, process, people, roles and responsibility setup, define maturity model, life cycle, operating model, training/guidelines, reference architecture, API governance and other necessary functions that suites bank needs.
   - **Key contributions:**
     a. Evaluated the current technology stack, development and CI/CD processes, API/service lifecycle, identified process gaps, and determined pain points, established both current-state and future-state specifications.
     b. Led workshops with subject matter experts (SMEs), client architects, and leaders for technology and application evaluations.
     c. Led the workshops on API Harvesting and Blueprinting process and guidelines using Domain Driven Design, TDD, BDD approach.
     d. Drove several proof-of-concept initiatives using Infosys internal accelerators to expedite the overall SDLC development cycle.
     e. Created an Organizational API adoption maturity model tailored to HSBC's needs.

8. **Next-Gen Investment Data Platform**:
   - **Client: Nationwide Insurance**
   - **Role:** Enterprise Solution Architect
   - **Technologies Used:** ETL, J2EE, SOA, Vendor Analysis (Corios, Novantas, Model Rick Group, Informa, Kantar Media), Decision Modelling, and Visio for Blueprint
   - **Project Role Description:** Re-architecture of existing data integration process (Informatica) to decrease process toll license cost and improve the long-change request process.
   - **Key contributions:**

    a. Restructured existing data processing platform with distributed, event-driven microservice architecture and open-source Java computing frameworks (Spring Cloud Data Flow).
    b. Implemented MVP based on new recommendations.

9. **Mortgage Banking (Loan Origination, Pre-Qual, Decision Engine, Digital Underwriting)**:
    - **Client: Capital One Financial Corporation**
    - **Role:** Enterprise Solution Architect
    - **Technologies Used:** Java, J2EE, SOA, RESTful API, Microservices Architecture, DevOps, Empower LOS, Drools 6.2/JBPM 6.2, OMG DMN 1.0 Specification, Swagger Specification, DocuSign, Alfresco, Kafka, Mongo DB, Amazon VPC, EC2, S3, Chef, Jenkins, and Maven
    - **Project Role Description:** Implementation of a system to enable business decisions to be encapsulated, maintained, deployed, and implemented independently of the application using the system.
    - **Key contributions:**
        a. Collaborated with the Business Experience Team in determining personas, customer journeys, and interaction for the ING-based existing loan origination.
        b. Restructured existing ING-based products, such as domain gateway (DGW) dependency removal; utilized REST-API-based solutions to the middle layer.
        c. Designed Home Loan "Cimplified Pricing Engine" as part of their home-loan pre-qualification workflow automation using OMG BPMN and DMN specification, runtime as Drools 6.2 and JBPM flow.

10. **Operation Decision Mngt. System Migration**:
    - **Client: KeyCorp**
    - **Role:** Enterprise Solution Architect
    - **Technologies Used:** Java, J2EE, SOA, IBM BPM/Lombardi, IBM WODM 8.0.0 BRMS, IBM DataPower, IBM WebSphere AppServer 7.0, RAD 7.5, and PVCS
    - **Project Role Description:** Reduction of data center footprint by restructuring the existing loan-origination and fulfilment, pre-qualify, loan calculator tools, and digital underwriting project to meet the new digital architecture specification.
    - **Key contributions:**
        a. Evaluated the IBM decision management system and IBM BlueworksLive, and centralized the BRMS architecture implementation to prevent IBM CPU-based license costs.
        b. Redesigned the key shared decision management system to support the new IBM WODM 8.0.x
        c. Defined guidelines for future migration steps on workflow and case management automation, rules harvesting methodology, maturity matrix, and roles and responsibility models.

11. **Key Intelligent Profile (KIP)**:
    - **Client: KeyCorp**
    - **Role:** Solution Architect
    - **Technologies Used:** JSF, JQuery, Java, J2EE, SOA, IBM DataPower, IBM-ILOG JRules 7.1.x BRMS, Siebel, IBM BlueworksLive, IBM BPM/Lombardi, IBM Websphere AppServer 7.0, RAD 7.5, and PVCS
    - **Project Role Description:** Implementation of the KIP aspect of the Keyvolution Business Transformation project, providing the salesforce 360-degree view of a customer in a relationship.
    - **Key contributions:**
        a. Led the master webservice specification design using approximately 50 independent services belongs to Demographics, Insights, Accounts, and Interactions Group.
        b. Integrated the customer stale-data and updation process workflow automation using IBM BPM, ILOG JRules technology.

c. Coordinated with business SMEs in determine existing issues and requirements for their business need.

12. **Risk Based Client Due Diligence Project**:
    - **Client: KeyCorp**
    - **Role:** Solution Architect
    - **Technologies Used:** Java, JSP-Struts, J2EE, SOA, IBM BlueworksLive, IBM BPM/Lombardi, IBM Filenet, ILOG JRules 6.6 BRMS, WAS 6.1, Oracle 10g, Cruise Control System, IBM Rational Application Developer, and PVCS
    - **Project Role Description:** Design and deployment of a solution to effectively identify, manage, and screen the bank's new and existing clients.
    - **Key contributions:**
        a. Designed and implemented highly scalable and resilient fraud and high-risk customer detection identification system and due diligence process automation and case management using the J2EE, SOA, and IBM BPM, and IBM Filenet.
        b. Designed to implement existing 3500+ business rules using ILOG JRules 6.6 centralized BRMS systems.
        c. Optimized the JVM configuration for deployment in WebSphere, enabling the system to support both batch and online processing of over 1 million customer information changes daily.
        d. Coordinated with business SMEs in determine existing issues and requirements for their business need ▪ Restructured the bank's customer risk profile identification and escalation modules supporting AntiMoney Laundering (AML) business compliance.

## EDUCATION AND CREDENTIALS

**Master of Business Administration (M.B.A.) - Systems Management** Jul 1999 – Jun 2001
University of Calcutta, Indian Institute of Social Welfare and Business Management, Kolkata

**Bachelor of Science (B.S.) – Economics, Statistics, Mathematics** Jul 1995 - Jun 1998
University of Calcutta, Kolkata

## LICENSES & CERTIFICATIONS

AWS Certified Solutions Architect – Professional 2018
AWS Certified Solutions Architect - Associate 2018
TOGAF 9 Certified 2018
Sun Certified Java Programmer (SCJP) 2003

## HONORS & AWARDS

- Infosys Awards for Excellence 2023
- Infosys Technology Awards 2020
- Architect of the Year by Cognizant 2016
- #1 Home Loans Digital Experience Journey by Capital One 2015
- Above & Beyond by Cognizant 2008, 2011
- Dedicated Commitment by KeyBank 2012
- Top 10 Software Engineer by Satyam 2004, 2005

## PUBLICATIONS

https://www.linkedin.com/pulse/api-discovery-have-fun-debasish-dalui/
https://www.infosys.com/services/api-economy/insights/organizational-maturity-model.html